

Julien ROYER

Artificial Intelligence Engineer

Double-degree engineer bridging Intelligent Systems Engineering and a research-oriented Master in Artificial Intelligence. Focused on the theoretical foundations of machine learning and their integration into complex industrial environments.

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PROFILE

Currently at **Thales Alenia Space**, working on the integration of advanced AI solutions into operational industrial processes in the aerospace sector. I hold a double degree in **Intelligent Systems Engineering** (ENSE3) and a **Master in Artificial Intelligence** (ENSIMAG) — a research-oriented program focused on the foundations and advanced methods of machine learning. Particularly interested in the theoretical foundations of ML, as well as their adaptation and deployment into complex industrial environments, including training optimization and integration into application pipelines.

Fields of interest: Agentic AI • Applied Machine Learning • Computer Vision

PROFESSIONAL EXPERIENCE

AI Engineer — Thales Alenia Space

Feb 2026 – Aug 2026

[Internship, Toulouse, France](#)

End-of-study internship

- Designed end-to-end AI solutions for Integration, Validation and Qualification (IVQ) processes in space systems.
- Implemented **Agentic Generative AI** pipelines for technical document analysis and generation, automated synthesis, and decision support.
- Solution **deployed across several engineering services and teams**; the backend was taken over by the AI department to be turned into a **finished industrial product**.
- Keywords: Generative AI, LLM Agents, Industrial AI, DevOps, Space Systems.

Research Intern — Numerical Methods, INRIA

Jun 2025 – Sep 2025

[Pau, France](#)

Makutu team (Inria / TotalEnergies)

- Analysed and implemented Discontinuous Galerkin (DG) formulations on pyramid-type meshes for acoustic wave propagation — a mathematically demanding problem at the frontier of numerical simulation.
- Conducted a literature review, derived the mathematical method variants, and selected the Bergot–Cohen–Duruffé nodal basis.
- Implemented the methods in **C++** and integrated them into the open-source HPC code **GEOS**; authored the full technical documentation.
- Keywords: Research, Applied Mathematics, HPC, Numerical Methods, Open Source.

Engineering Intern — SOFRESID Engineering

May 2023 – Jun 2023

[Pau, France](#)

Chemical industry

- Wrote calculation methods and sizing programs for safety-valve engineering, applied to live chemical industry projects.
- Performed process simulations using ASPEN HYSYS; first exposure to real industrial constraints and norms.

EDUCATION

Master of Artificial Intelligence (M2) — Grenoble INP, ENSIMAG

2025 – 2026

[Foundations and Advanced Methods in Machine Learning](#)

- Mathematical foundations of Machine Learning, GPU Computing, Generative AI, Mathematical Optimisation, AI Ethics.

Engineering Degree — Grenoble INP, ENSE3

2023 – 2026

[Automation & Intelligent Systems Engineering track](#)

- Cyber-physical systems control & supervision, Signal Processing, performance analysis and optimisation, engineering project management.

International Exchange Semester — McMaster University

2025

[Hamilton, Ontario, Canada](#)

- Object-Oriented Programming, Modelling & Numerical Solutions, Advanced Control Theory, Electronics & Instrumentation, Marketing.

Preparatory Classes — La Prépa des INP

2021 – 2023

- Mathematics, Physics and Engineering Science — foundation for the competitive French engineering entrance exams.

SELECTED PROJECTS & RESEARCH

KYM — Multi-Agent System for Knowledge Retrieval, Synthesis & Generation

Thales Alenia Space

- Proprietary multi-agent system for complex engineering tasks over dense aerospace and industrial documentation.
- **Technical core:** agentic loop for task decomposition and tool execution, with an advanced hybrid RAG pipeline (Multi-Query + HyDE expansion, parallel vector and BM25 search, RRF fusion, final reranking).
- **Other use cases:** automated technical documentation generation, and a failure-segmentation agent parsing logs to isolate complex system faults.

Discontinuous Galerkin Methods on Pyramid Meshes

INRIA / GEOS

- Research implementation of DG formulations on pyramid-type elements: derivation, C++ integration into the open-source GEOS simulator, and full technical documentation.

GPU-Accelerated Particle Simulation

- GPU-accelerated 2D particle simulator (CUDA) handling thousands to millions of interacting particles in real time for fluid simulation, based on a personal C++ Verlet-integration simulator; benchmarked CPU vs. GPU with shared-memory and spatial-hash-grid optimisations.

Performance Analysis of a Tiny Diffusion Model

- Performance study of a minimal PyTorch DDPM implementation, combining Python- and system-level profiling (cProfile, Hyperfine) with framework-level profiling (PyTorch Profiler) to identify training bottlenecks and derive optimisation insights.

Deep Q-Learning Agent for Sailing Navigation

Reinforcement Learning project

- Design, implementation and training of a Deep Q-Network (DQN) agent for autonomous sailing navigation in dynamic wind fields, with feature engineering, experience replay and detailed performance analysis.

TECHNICAL SKILLS & LANGUAGES

AI / ML	Agentic systems, RAG, LLM fine-tuning, training & inference optimization, deep learning, generative AI, computer vision, reinforcement learning	French	Native
Scientific Computing	Python, PyTorch, DeepSpeed, NumPy, Matlab	English	Fluent — C1
HPC / Low-level	C, C++, CUDA, OpenMP, MPI, performance profiling		Cambridge
Software & Dev	Software architecture, Docker, CI/CD, databases (SQL / vector DB), REST APIs, Git	Spanish	Advanced — C1
Infra	Linux, Bash, cluster, deployment pipelines	Interests:	Rugby (regional & national level), mountain hiking, oenology.